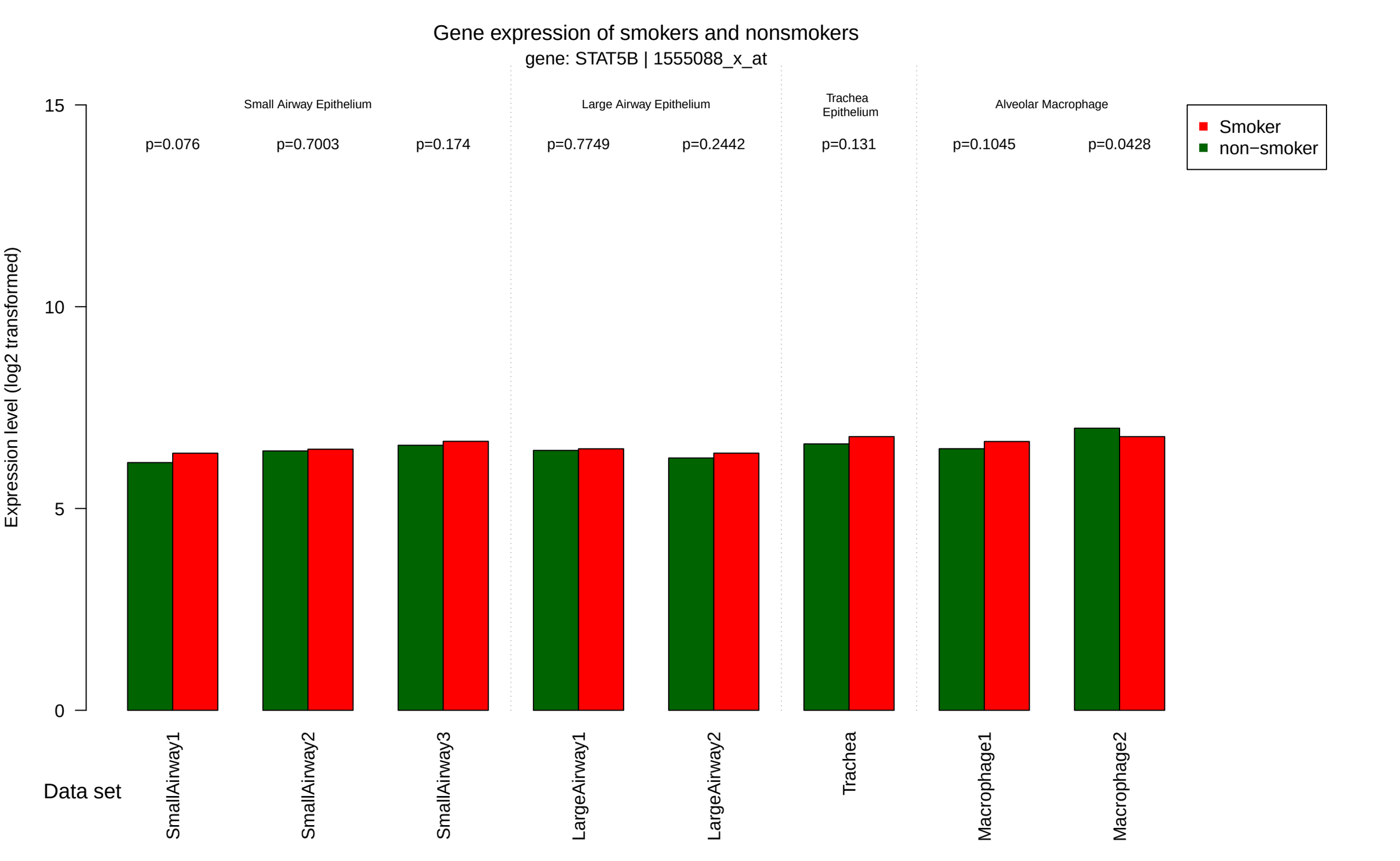
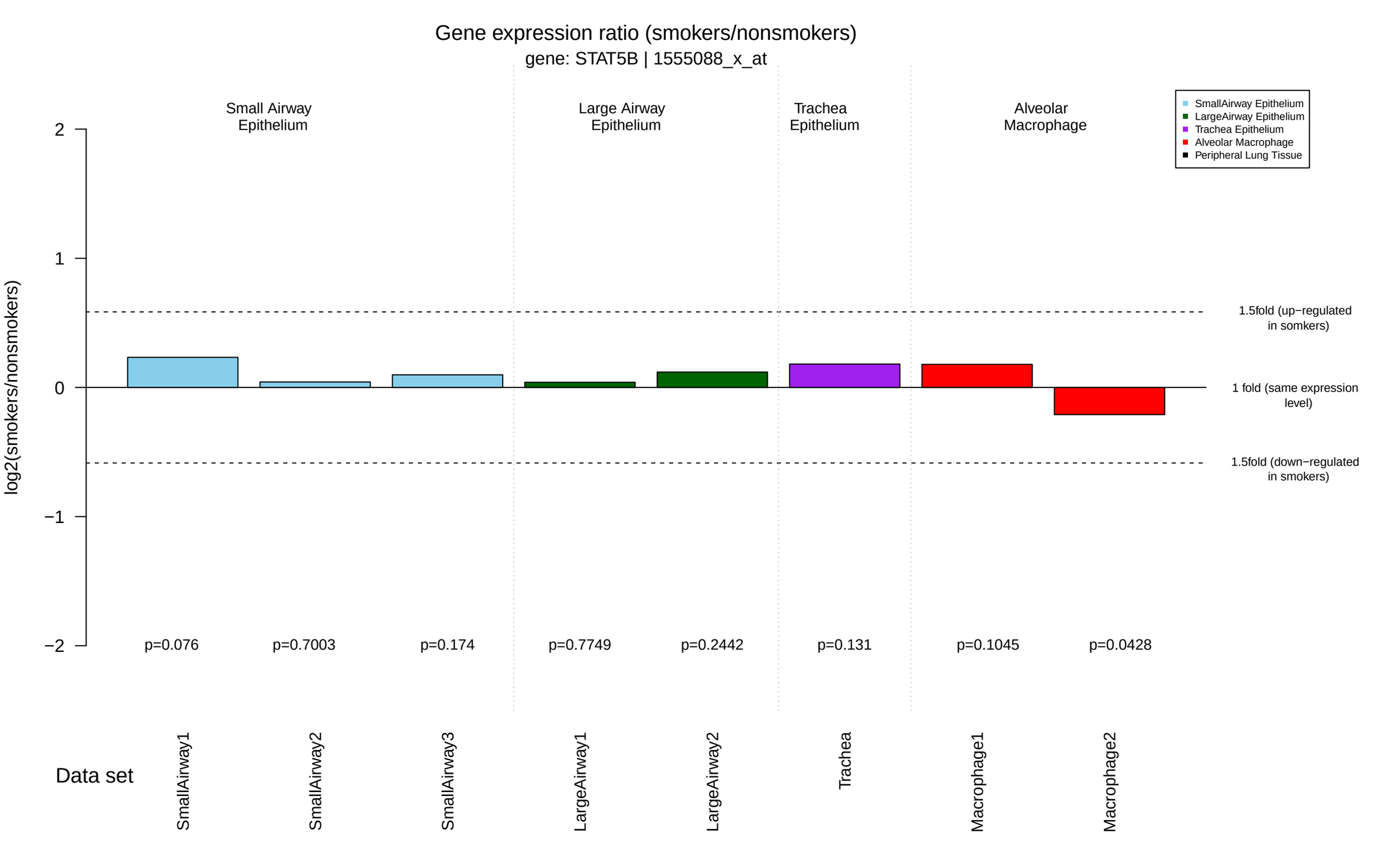
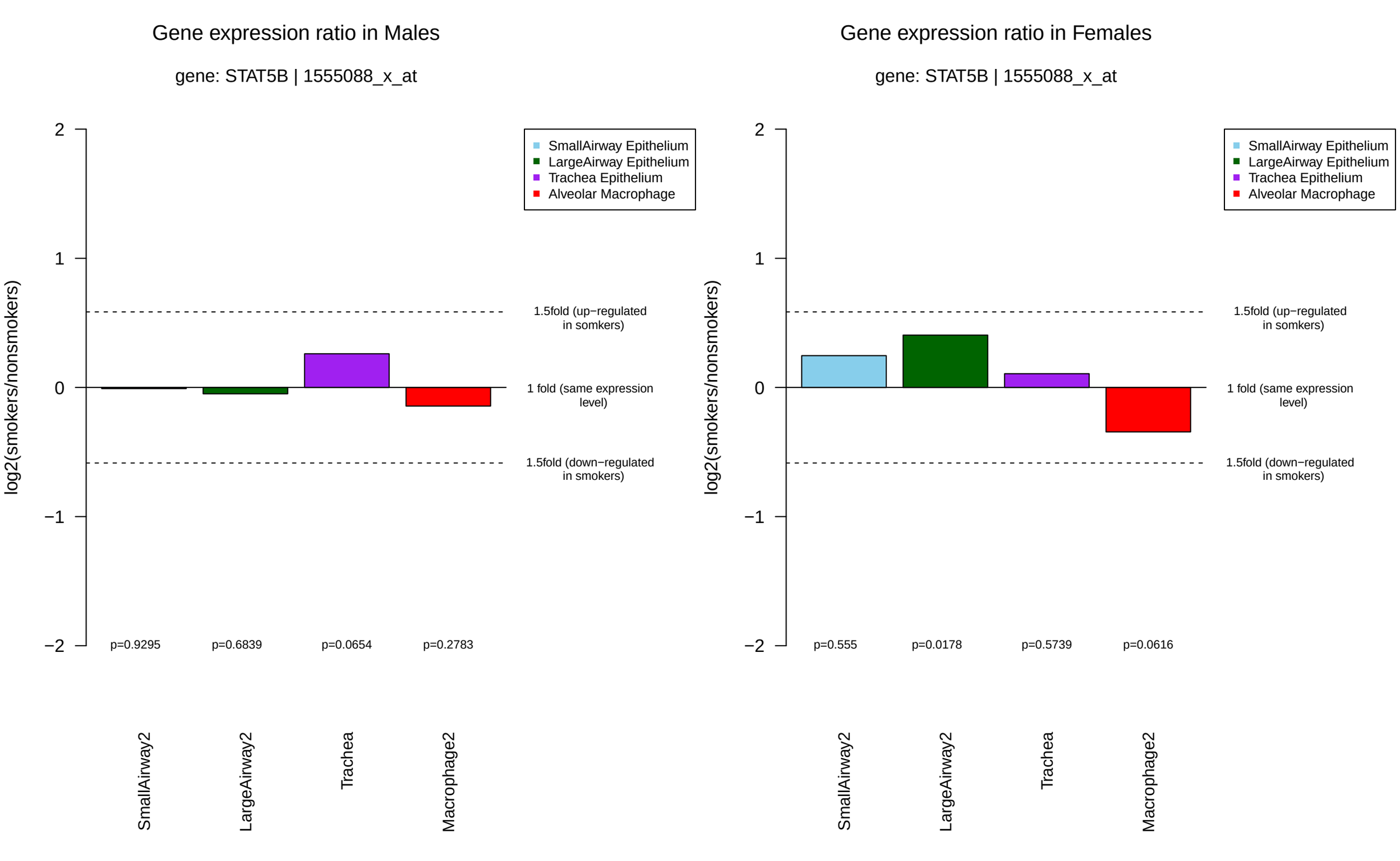
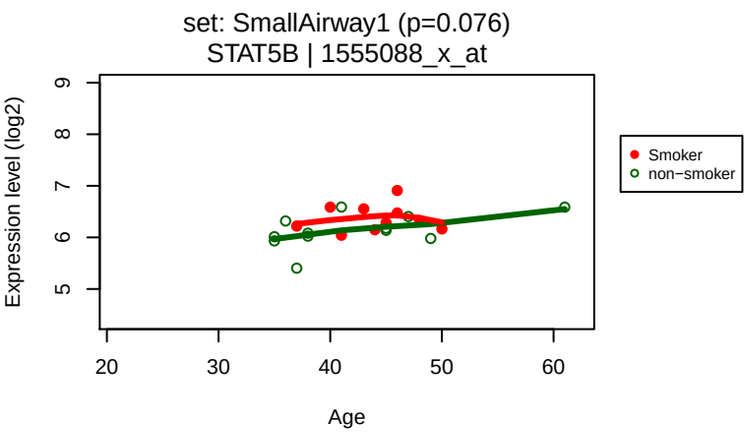
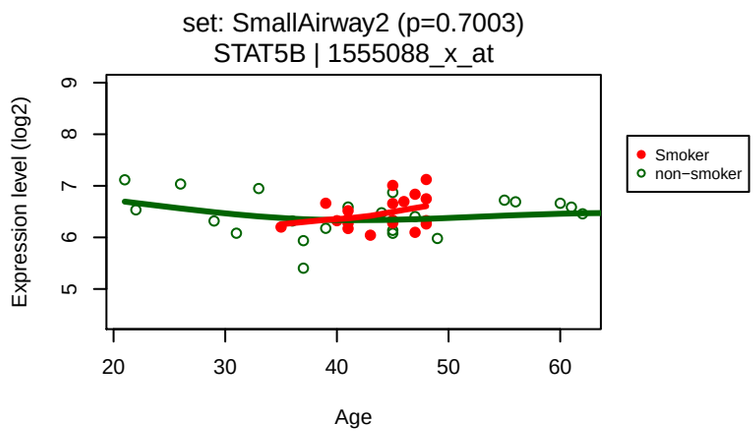
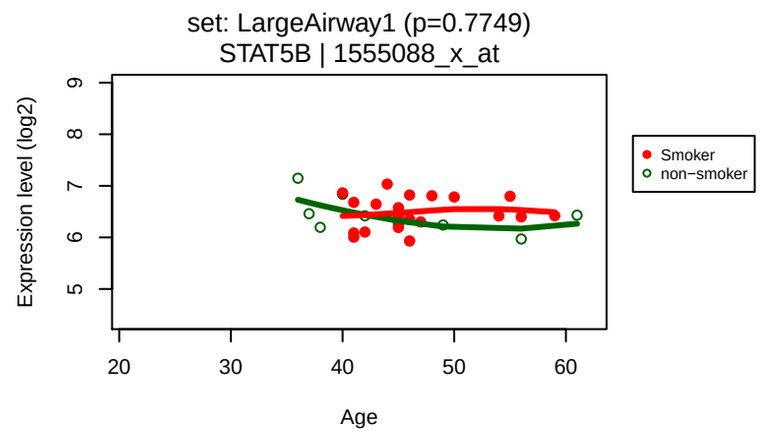

Gene expression of smokers and nonsmokers
gene: STAT5B | 1555088_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.076
p=0.7003
p=0.174
p=0.7749
p=0.2442
p=0.131
p=0.1045
p=0.0428
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

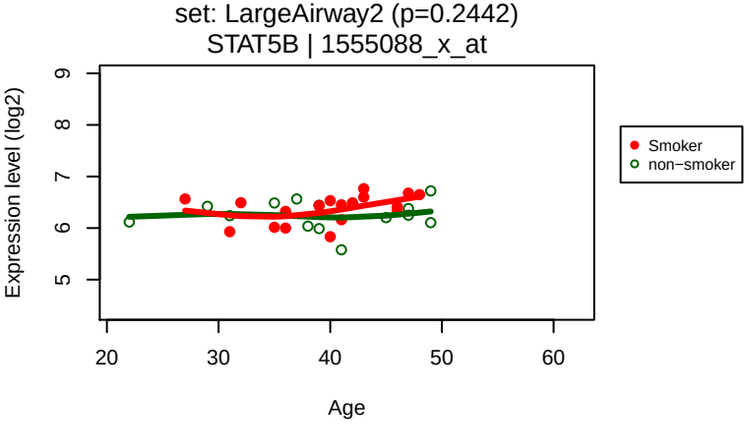

Gene expression ratio (smokers/nonsmokers)
gene: STAT5B | 1555088_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.076
p=0.7003
p=0.174
p=0.7749
p=0.2442
p=0.131
p=0.1045
p=0.0428
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

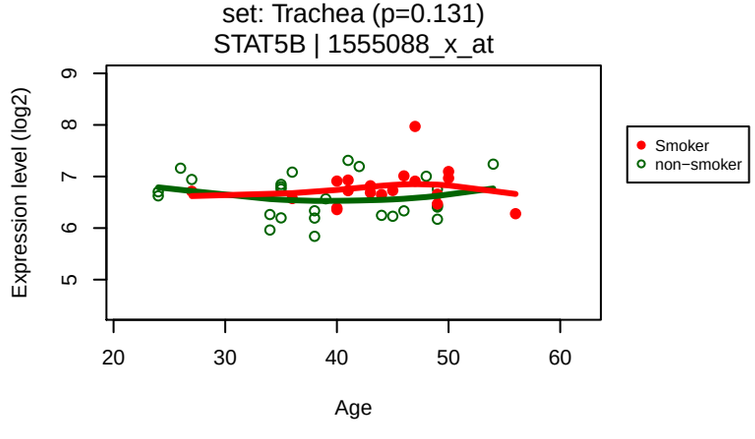

Gene expression ratio in Males
gene: STAT5B | 1555088_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9295
p=0.6839
p=0.0654
p=0.2783
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STAT5B | 1555088_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.555
p=0.0178
p=0.5739
p=0.0616
SmallAirway2
LargeAirway2
Trachea
Macrophage2

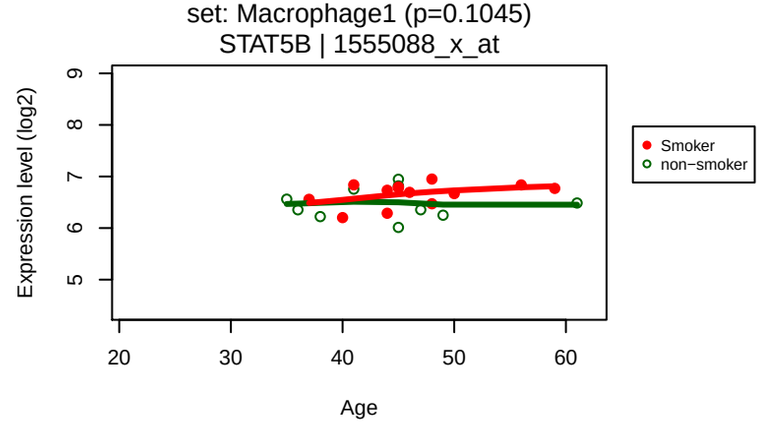

set: SmallAirway1 (p=0.076)
STAT5B | 1555088_x_at
Expression level (log2)
Age
Smoker
non−smoker

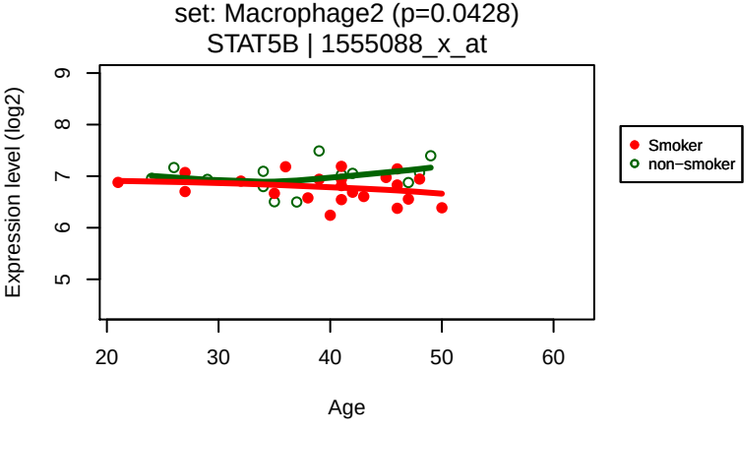

set: SmallAirway2 (p=0.7003)
STAT5B | 1555088_x_at
Expression level (log2)
Age
Smoker
non−smoker

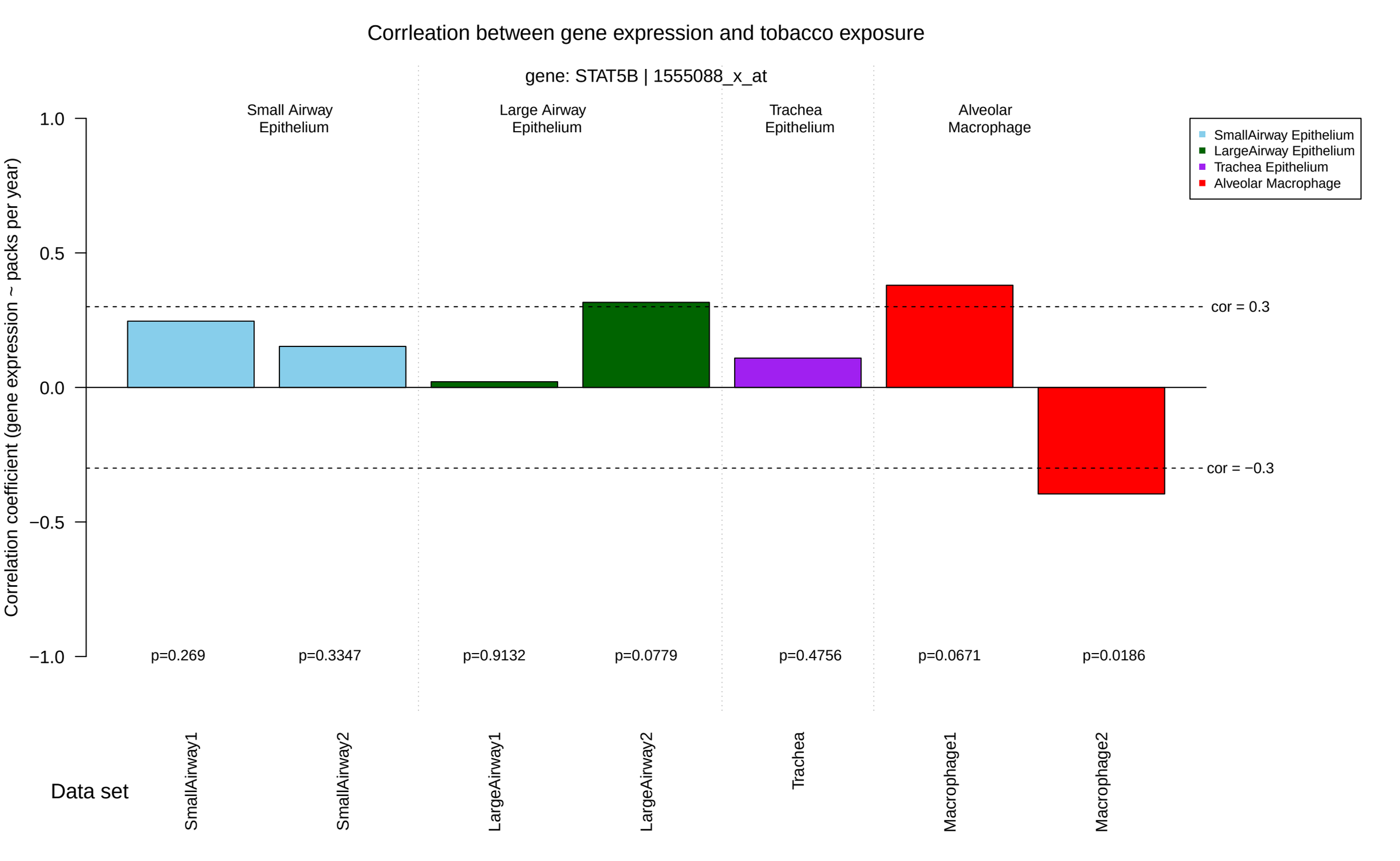

set: LargeAirway1 (p=0.7749)
STAT5B | 1555088_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2442)
STAT5B | 1555088_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.131)
STAT5B | 1555088_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1045)
STAT5B | 1555088_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0428)
STAT5B | 1555088_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STAT5B | 1555088_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.269
p=0.3347
p=0.9132
p=0.0779
p=0.4756
p=0.0671
p=0.0186
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2